HONGXIANG ZHANG

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RESEARCH INTERESTS

My research focuses on reliable and efficient AI systems, with an emphasis on LLM reasoning, multi-agent communication, hallucination mitigation, and AI for code. I am broadly interested in developing robust agentic workflows by combining alignment, decoding, and lightweight diagnostic methods across software engineering, security, and generative AI applications.

EDUCATION

Purdue University — August 2025 - Present
Ph.D in Computer Science; Advised by Prof. Tianyi Zhang

University of California, Davis — Sep 2022 - June 2025
M.Sc. in Computer Science

Australian National University — Jul 2020 - Sep 2022
B.Eng. (Honours) in Computer Science

Shandong University — Sep 2018 - Jun 2022
B.Eng. in Computer Science

ACADEMIC EXPERIENCE

Enhancing Multi-Agent Communication through Attention Steering with Context Relevance
Hongxiang Zhang, *Yuan Tian, Tianyi Zhang* — [arXiv]
[Under review] | Multi-agent systems; Attention Alignment; Context Management

· Proposed a training-free context management method that steers each agent's attention toward relevant conversation history using temporal and spatial decay.
· Improved multi-agent performance across five benchmarks by up to 7.64 absolute points while remaining robust as the number of agents and interaction rounds increases.

FLARE: Fine-Grained Diagnostic Feedback for LLM Code Refinement
Yinsheng Yao, ***Hongxiang Zhang****, Weixi Tong, Tianyi Zhang* — [arXiv]
[Under review] | Code Refinement; Fault Localization; Test-time Scaling

· Proposed an iterative code-refinement framework centered on a lightweight fault localizer that predicts line-level suspiciousness for targeted LLM refinement.
· Improved LiveCodeBench and BigCodeBench performance across five LLMs, with up to 7.42% gain without candidate search and 8.50% average gain with top-k search.

Attention-Aligned Reasoning for Large Language Models
Hongxiang Zhang, *Yuan Tian, Tianyi Zhang* — [arXiv]
[Under review] | LLM Reasoning; Attention Alignment; Prompt Engineering

· Proposed a reasoning-time alignment method that uses step-by-step attention alignment to keep LLMs focused on the prompt and intermediate goals.
· Outperformed existing prompting-based approaches and RL-trained SLMs.

Active Layer-Contrastive Decoding Reduces Hallucination in Large Language Model Generation
Hongxiang Zhang, *Hao Chen, Muhao Chen, Tianyi Zhang* — [arXiv]
[EMNLP 2025 Main] | Hallucination Mitigation; Decoding; Reinforcement Learning

- Proposed an RL-guided decoding strategy that actively contrasts model layers to suppress hallucinations.
- Achieved state-of-the-art hallucination reduction across five generation benchmarks.

LLAMAFUZZ: Large Language Model Enhanced Greybox Fuzzing
Hongxiang Zhang, *Yuyang Rong, Yifeng He, Hao Chen* [arXiv]
[AST 2026] | LLM for Systems; Greybox Fuzzing; Software Testing

- Built LLAMAFUZZ, an LLM-enhanced greybox fuzzer that learns structured input formats and mutation strategies for binary and text targets.
- Improved real-world coverage and bug finding, identifying 47 unique bugs and increasing branch coverage by 27.19% over AFL++.

SteerDiff: Steering towards Safe Text-to-Image Diffusion Models
Hongxiang Zhang, *Yifeng He, Hao Chen* [arXiv]
Safe Generative AI; Prompt Steering; Diffusion Models

- Developed **SteerDiff**, a lightweight adapter that steers prompt embeddings away from unsafe concepts before text-to-image diffusion.
- Preserved generation quality while improving robustness against red-teaming and supporting concept forgetting tasks.

TEACHING

ECS 153 Computer Security - UC Davis	2024 Spring
ECS 036A Programming & Problem Solving - UC Davis	2024 Winter
ECS 036C Data Structures, Algorithms, & Programming - UC Davis	2023 Fall
ECS 140A Programming Language - UC Davis	2022 Winter
ENGN 4528 Computer Vision - Australian National University	2021

INDUSTRY EXPERIENCE

Software and Operation Engineer Intern Volkswagen, Beijing, China Feb 2022 - Jul 2022

- Achieved data masking and auto-populated to **JIRA** log in Python.
- Intuitively analyzed and provided services and finance data to CIO with **Tableau**, helping the management level adjust the company service strategy agilely.

Quality Assurance Intern Didi Global, Beijing, China Dec 2020 - Feb 2021

- Developed distributed Java invoice services to enable the interaction between server and end-user devices that served more than 1 million users per day.
- Achieved unit testing automation by using **JUnit** and the test case in **Redis**, reduced over 15% of the testing engineer's workload.

HONORS AND AWARDS

Microsoft Certified: Azure Fundamentals	*2022*
Shandong University Scholarship	*2018- 2019*
Shandong University Research and Innovation Specialty Scholarship	*2018- 2019*
Lan Qiao Collegiate Cup Provincial Third Award	*2019*
Shandong University Scholarship	*2020- 2021*
Excellent Course Representative of ENGN4528/6528	*2021*